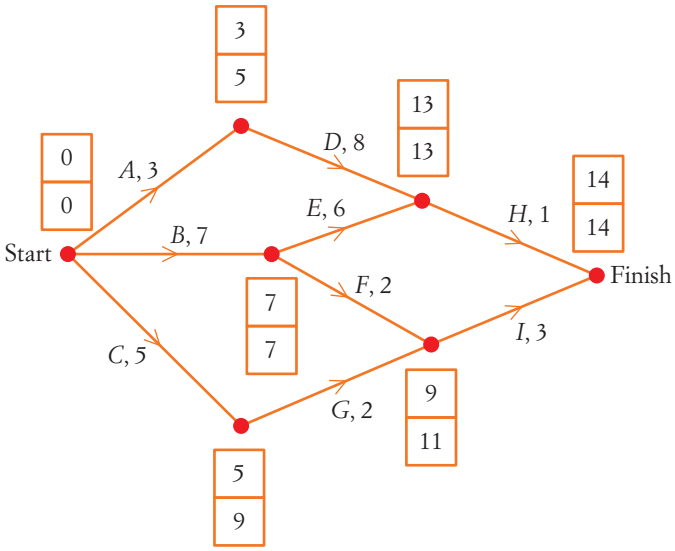
WORKSHEET

Chapter 9: Critical paths, critical times and activity float times

1 The earliest and latest starting times have been calculated for the network below. Times shown are in hours.



a Find the critical path/s and critical time for the project.

b Calculate the float times for activity:

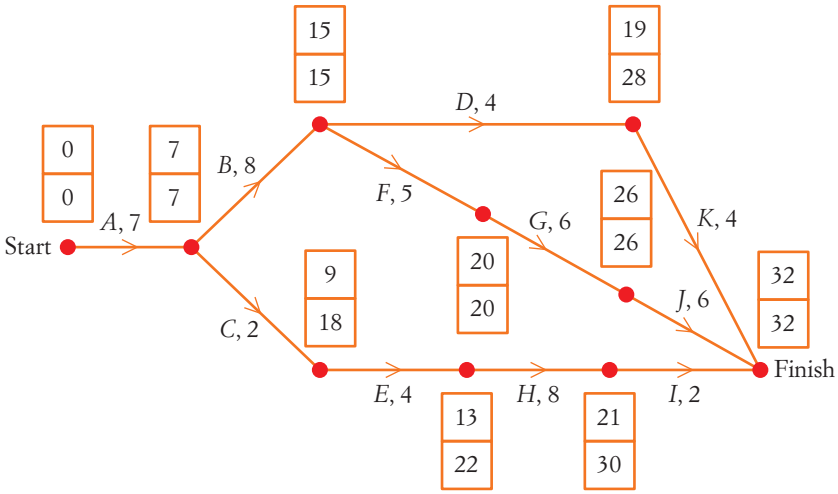
i A

ii G

iii H.

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2 The EST and LST have been calculated for each activity in the project below. Times shown are in days.



a Find the critical path/s and critical time for the project.

b Calculate the float times for activity:

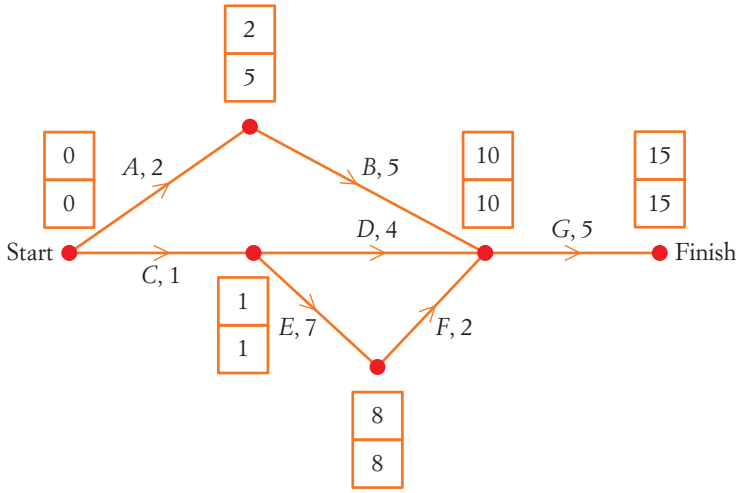
i F

ii H

iii K.

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3 The network for a project is shown. Times shown are in days. The EST and LST have been calculated for each activity.



a Determine the critical path/s and critical time.

b Determine the float time for activity *B*.

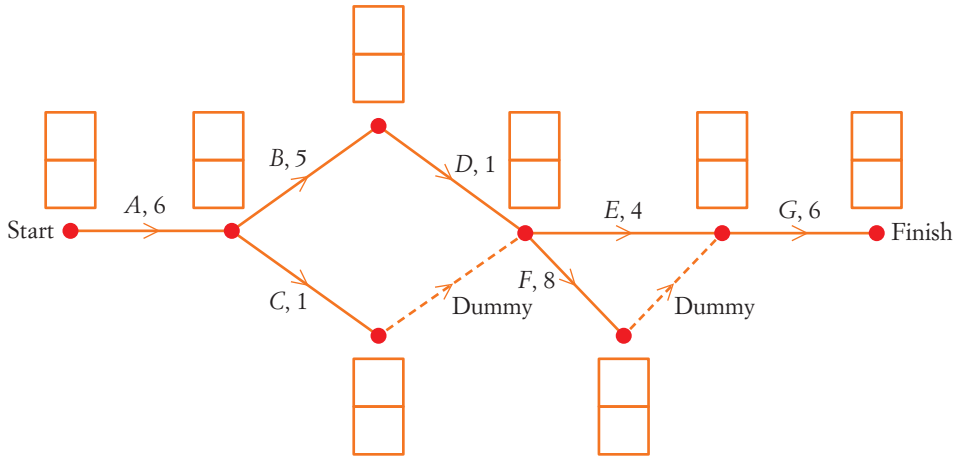
c If the time taken to complete activity *F* is reduced to 1 hour, state:

i the critical path/s

ii the critical time.

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4 The network for a project is shown. Times shown are in hours.



a Forward scan to determine the earliest start times for each activity.

b Backward scan to determine the latest start times for each activity.

c Determine the critical path/s.

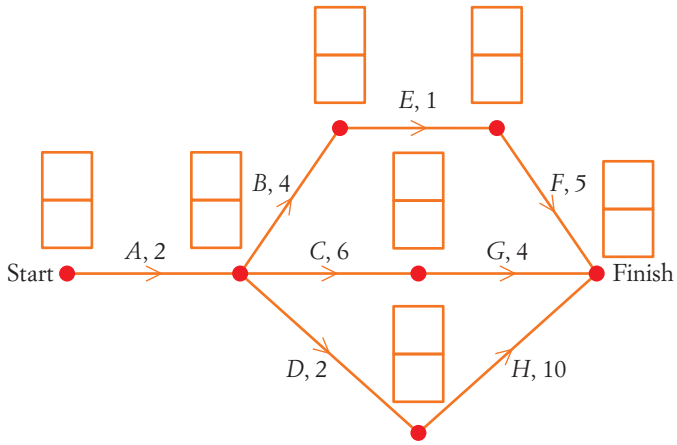
d If the time taken to complete activity *B* is reduced by 4 hours, state:

i the critical path/s

ii the critical time.

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5 In the network below, times are shown in days.



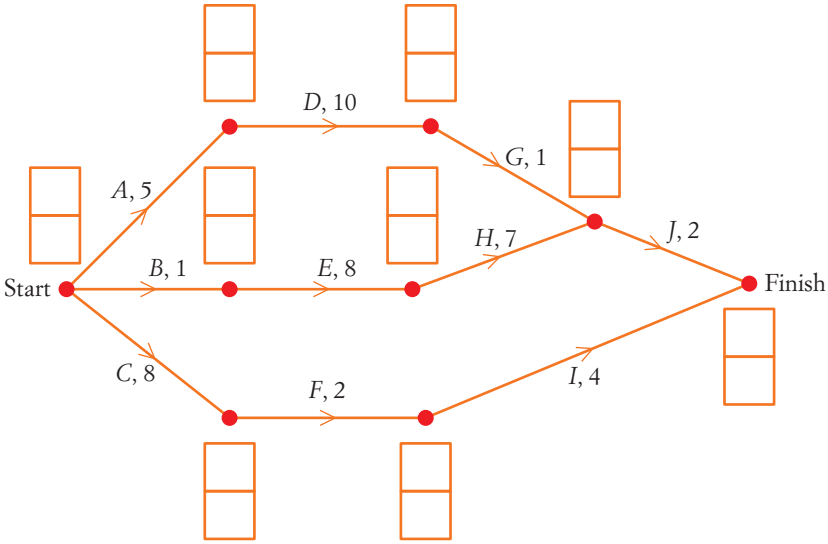
a Use the network diagram to find the EST and LST for each activity in the network.

b Determine the critical path/s.

c Determine the float times for all the non-critical activities in the network.

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6 In the network below, times are shown in days.



a Use the network diagram to find the EST and LST for each activity in the network.

b Find the critical path/s.

c Find the critical time.

d There is a delay of 11 days in activity *F*.

i Determine the new critical path/s.

ii Determine the new critical time.

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7 **a** Use the activity table below to construct a network diagram.

Activity	Time (days)	Immediate predecessor/s
A	15	None
B	7	None
C	20	None
D	21	*C*
E	18	*B*
F	9	*A*
G	14	*F*
H	2	*D*, *E*

b Calculate the EST and LST for each activity.

c Determine critical path/s.

d Determine the earliest completion time for the project.

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8 a Use the activity table below to construct a network diagram.

Activity	Time (hours)	Immediate predecessor/s
A	5	None
B	1	None
C	12	*B*
D	2	*A*
E	6	*A*
F	4	*C*, *E*
G	1	*B*
H	2	*G*
I	5	*D*
J	6	*F*
K	2	*H*

b Calculate the EST and LST for each activity.

c Determine the critical path/s.

d Determine the earliest completion time for the project.

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e There is a delay of 11 hours in activity *D*.

i Determine the new critical path/s.

ii Determine the new critical time.

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Answers

1 **a** *BEH*, 14 hours

b **i** 2 hours

ii 4 hours

iii 0

2 **a** *ABFGJ*, 32 days

b **i** 0

ii 9 days

iii 0

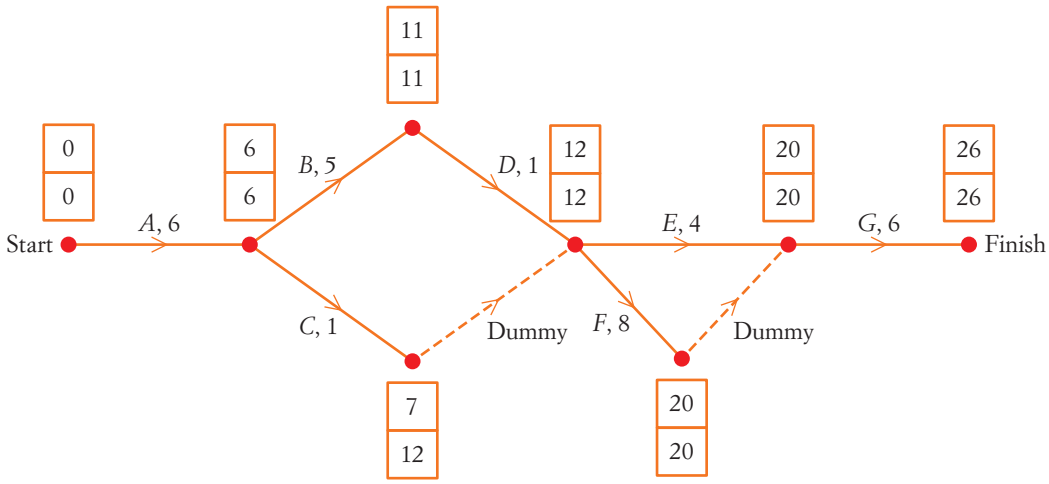
3 **a** *CEFG*, 15 days

b 3 days

c **i** *CEFG*

ii 14 days

4 **a, b**



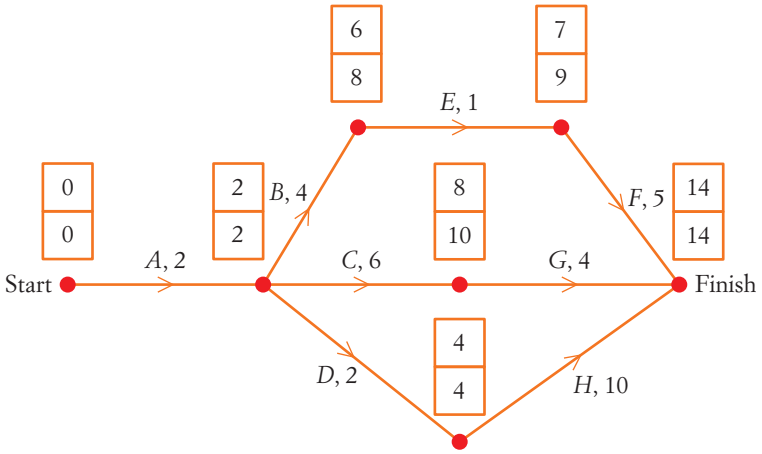
c *ABDFG*

d **i** *ABDFG*, *ACFG*

ii 22 hours

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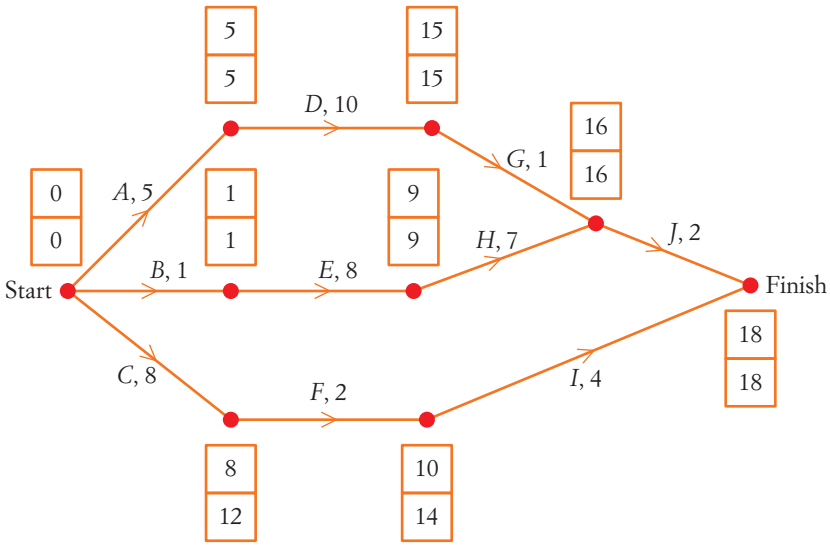
5 a



b *ADH*

c *B*: 2 days; *E*: 2 days; *F* 2 days; *C*: 2 days; *G*: 2 days.

6 a



b *ADGJ*, *BEHJ*

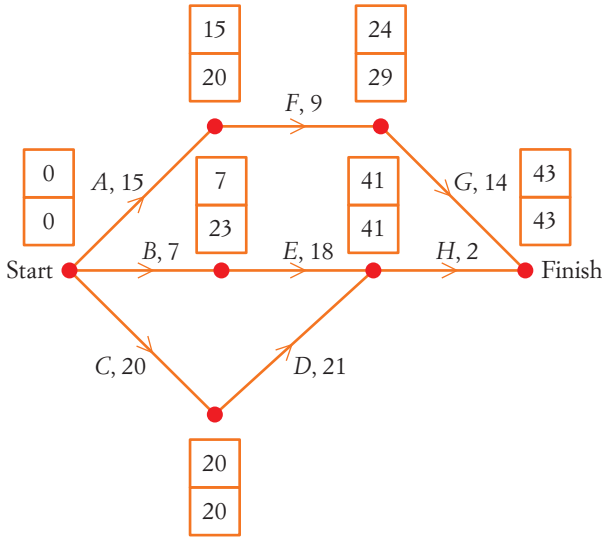
c 18 days

d i *CFI*

ii 25 days

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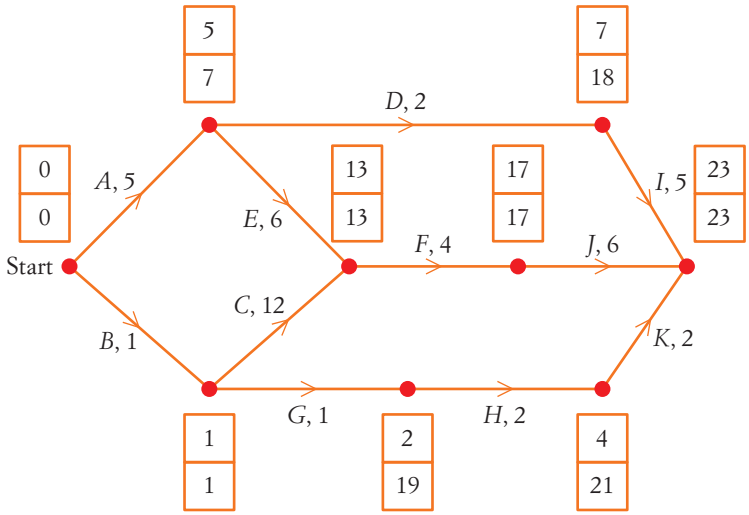
7 a, b



c *CDH*

d 43 days

8 a, b



c *BCFJ*

d 23 hours

e i *ADI*, *BCFJ*

ii 25 hours

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